

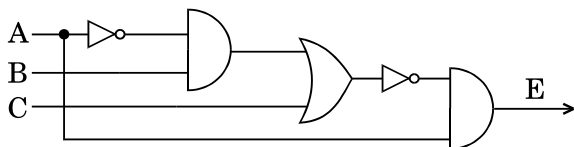
MCA (Revised) / BCA (Revised)**Term-End Examination****June, 2021****MCS-013 : DISCRETE MATHEMATICS***Time : 2 hours**Maximum Marks : 50*

Note : Question no. 1 is **compulsory**. Attempt any **three** questions from the rest.

1. (a) Write the truth value of the conjunction of :
“The earth is round” and “ $3 > 4$ ”. 2
- (b) Use Mathematical Induction to prove that :
$$1 + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{n^2} \leq 2 - \frac{1}{n} \quad \forall n \in \mathbb{N}. \quad 4$$
- (c) If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function such that
 $f(x) = 3x - 2$, prove that f is injective. Also
find the inverse of f . 5
- (d) Show that $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$
are logically equivalent. 3

- (e) A and B are two mutually exclusive events such that $P(A) = 0.4$ and $P(B) = 0.2$. What is the probability that :
- A does not occur ?
 - A or B does not occur ?
 - Either A or B does not occur ? 3
- (f) Find the number of ways of placing n people in $n - 1$ rooms, no room being empty. 3

- 2.** (a) What is integer partition ? Write down all the partitions of 8. Also find P_8^4 and P_8^7 . 4
- (b) Find Boolean Expression for the following logical circuit : 4



- (c) Let two functions be such that $f(x) = x^2 + 2$ and $g(x) = 2x$. Define fog and gof . 2
- 3.** (a) Reduce the following Boolean Expression to simpler form : 4
- $$E(X_1, X_2, X_3) = (X_1 \wedge X_2 \wedge X_3) \vee (X_1 \wedge X_2) \vee (X_2 \wedge X_3)$$
- (b) Show that $\sim(p \rightarrow q) \rightarrow p$ is a tautology. 2
- (c) Prove that $\sqrt{2}$ is irrational. 4

4. (a) What is Relation ? How is relation different from function ? Explain any two properties of relations with an example. 5
- (b) A company has the following professionals :
Project Leaders – 5, Team Leaders – 6,
System Architects – 3.
- Find how many different committees can be formed of 10 professionals, each containing at least 2 Project Leaders, at least 3 Team Leaders and at least 1 System Architect. 3
- (c) Find the dual of $A \cup B \cap C$. 2
5. (a) Explain the Identity Laws of Boolean Algebra. 2
- (b) State and prove the Addition Theorem of Probability. 4
- (c) Verify that $p \wedge q \wedge \sim p$ is a contradiction. 2
- (d) What is Exclusive Disjunction ? Write truth table for $p \oplus q$. 2
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